Jeongah Jasmine Lee

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Research Interests

Human-ai-vis · **Human-Computer Interaction** · **Human-AI Interaction** · **Visualization** · **Health & Well-Being Technologies**

I am a **human-AI interaction researcher** who builds and evaluates AI systems for health, well-being, and visualization through **mixed-methods research**. My work explores how **generative AI** can be designed to be transparent, safe, and genuinely helpful in everyday life, with a focus on **caregivers** and **people with disabilities**.

Education

MS/PhD	**University of Massachusetts Amherst**, Computer Science	Sep 2024 - 2029 (expected)
BS	**Sungkyunkwan University**, Computer Science and Engineering	Mar 2019 - Feb 2024
	The University of Texas at Austin (Exchange Student Program)	Jan - May 2022

Publications

Peer-Reviewed Conference and Journal

1. **RubRIX: Rubric-Driven Risk Mitigation in Caregiver-AI Interactions**
   - Drishti Goel, ***Jeongah Lee***, Qiuyue Joy Zhong, Violeta J. Rodriguez, Daniel S. Brown, Ravi Karkar, Dong Whi Yoo, Koustuv Saha
   - 2026 Findings of the Association for Computational Linguistics (ACL)
2. **From Text to Visuals: Using LLMs to Generate Math Diagrams with Vector Graphics**
   - Jaewook Lee, ***Jeongah Lee***, Wanyong Feng, Andrew Lan
   - AIED 2025: 26th International Conference on Artificial Intelligence in Education
3. **IoT Edge-Cloud: An Internet-of-Things Edge-Empowered Cloud System for Device Management in Smart Spaces**
   - Yoseop Joseph Ahn, Minje Kim, ***Jeongah Lee***, Yiwen Shen, Jaehoon Paul Jeong
   - 2023 IEEE Network Magazine

Preprints and Under Review

1–5: ACM CHI 2027 (under review) · 6–7: ACL (under review)

1. **Can LLMs identify and repair ruptures? Comparison between clinician practices and LLM behaviors**
   - ***Jeongah Lee***, Qiuyue Joy Zhong, Drishti Goel, Violeta J. Rodriguez, Dong Whi Yoo, Koustuv Saha, Ravi Karkar
2. **Conversational AI for Floorplan Accessibility: How Blind and Low-Vision Users Seek Spatial Information Through Dialogue**
   - ***Jeongah Lee***, Srikiran Kavuri, Hari Iyer, Ali Sarvghad
3. **Cohort-based Semantic Labeling: AI-Enabled Recovery of Visualization Semantics from Deployed SVGs**
   - ***Jeongah Lee***, Hima Varshini Surisetty, Durga Nirmaleswaran, Jahnavi Sharma, Srikiran Kavuri, Narges Mahyar, Ali Sarvghad
4. **Beyond Counting Blessings: Tracing the Evolution of Gratitude Practices and Technology Needs**
   - Qiuyue Joy Zhong, ***Jeongah Lee***, Drishti Goel, Violeta J. Rodriguez, Dong Whi Yoo, Koustuv Saha, Ravi Karkar
5. **En3: Head-Confined Multimodal Sensing for Engagement, Grounded in How Learners Define It**
   - Sidharth Anupkrishnan, Itir Sayar, ***Jeongah Lee***, Sri Harsha Musunuri, Guan-Ming Su, Tam Vu, Madeline Endres, Ravi Karkar, Phuc Nguyen
6. **Inform, Coach, Relate, Listen: Auditing LLM Caregiving Support Responses**
   - Drishti Goel, Agam Goyal, Veda Duddu, Olivia Pal, ***Jeongah Lee***, Qiuyue Joy Zhong, Violeta J. Rodriguez, Daniel S. Brown, Dong Whi Yoo, Ravi Karkar, Koustuv Saha
7. **Risk and Refusal: Mechanisms of LLM Responses to Suicidal Ideation**
   - Drishti Goel, Soorya Ram Shimgekar, ***Jeongah Lee***, Violeta J. Rodriguez, Daniel S. Brown, Ravi Karkar, Koustuv Saha

Research & Work Experiences

Dolby Advanced Technology Group, Research Intern — ***Atlanta, GA, USA*** — May - Aug 2026
- Advisor: Dr. Gautam Bhattacharya, Dr. Feiyu Lu, Dr. Jian Chen, Dr. Tica Lin
- Conducted research on designing proactive AI agents for spatial audio mixing

WHAT lab @ University of Massachusetts Amherst, Graduate Research Assistant — ***Amherst, MA, USA*** — Sep 2024 - Present
- Advisor: Dr. Ravi Karkar
- Led research on AI-driven mental health tools for Alzheimer's caregivers, integrating journaling,

conversational interaction, and cognitive reframing (CBT) support

Seoul National University Bundang Hospital, Natural Language Processing (NLP) Researcher — ***South Korea*** — Mar - Jul 2024
- Developed a predictive model for lung cancer TNM staging using Electronic Health Record (EHR) datasets, improving Macro F1-score from 0.70 to 0.93 through prompt design and data augmentation

Cipherome, Inc., Machine Learning (ML) Engineer Intern — ***San Jose, CA, USA*** — Mar - Aug 2023
- Developed the pipeline for an ML module within a clinician-focused medical data analysis platform
- Designed Figma wireframes to improve the platform's user interface and experience (UI/UX)

IoT lab @ Sungkyunkwan University, Undergraduate Student Researcher — ***South Korea*** — Jul - Nov 2021
- Advisor: Dr. Jaehoon Paul Jeong
- Developed an IoT edge-empowered Cloud System for the visual control of IoT devices in a user's smartphone, improving location tracking accuracy by 39% through the integration of Smart Pedestrian Dead-Reckoning and Particle Filter-Indoor Positioning algorithm

SK Planet Co.,Ltd., Industry-Academic Cooperation Student Researcher — ***South Korea*** — Mar - Dec 2020
- Developed an AR rhythm game application for the Busan One Asia K-pop Festival using AR Core to provide location-based interactive experiences

Awards & Honors

[Awards]
- 3rd Place, Chung-ang University AI and Humanities Academic Paper contest — 2023
- 1st Place, Kookmin University self-driving contest — 2021
- 3rd Place, Sungkyunkwan University AI x Bookathon contest — 2021
- Volunteering Excellence Prize, NIA (National Information Society Agency) — 2020

[Scholarships]
- Grace Hopper Conference Scholarship — 2026
- Academic Excellence Scholarship (top 12%) — 2022
- Creative Scholarship (100% tuition support) — 2021
- Sungkyun Software Scholarship (100% tuition support) — 2019
- MegastudyEdu Scholarship (external) — 2019

Teaching & Mentoring

[Teaching]

Graduate Teaching Assistant — Sep 2024 - Dec 2025
- *CICS 110 Introduction to Programming (Fall 2024)*: Led two lab sessions (60 students each) with hands-on instruction and grading support.
- *CS 383 Artificial Intelligence (Spring 2025)*: Mentored students in 15 group projects and assisted with assignment design and evaluation (180 students).
- *CS 571 Data Visualization and Exploration (Summer 2025), INFO 348 Data Analytics with Python (Fall 2025)*: Sole TA for CS 571 (35 students) and INFO 348 (60 students); independently managed grading, office hours, and assignment design.

[Mentoring]

Undergraduate Research Volunteer Program Mentor — May - Sep 2025
- Gerindra Adi, Vidhaan Kothari, Nish Methuku (CS Undergraduate)
- Topic: Glanceable Health Visualization for Older Adults Using Large Multimodal Models

Services

IEEE VIS Poster Committee — 2026

EuroVis Reviewer — 2026

Skills

Technical Proficiencies: Python, Pytorch, JavaScript (React.js, D3.js, Node.js), Flask, HTML, CSS, MongoDB
Research Methodologies:
Qualitative: User Research, Semi-Structured Interviews, Prototyping, Usability Testing, Think-Aloud Protocol, Thematic Analysis
Quantitative: Survey Design, Experimental Design, Statistical Analysis, Modeling, Visualization Analysis
Tools & Platforms: Git, GitHub, Vercel, Render, Capacitor, Docker, Figma, Zeplin